How Retrieval-Augmented Generation Works

Retrieval-augmented generation, or RAG, answers questions using a set of documents. A language model on its own answers from what it learned during training, so it can be out of date and knows nothing about your private files. RAG adds a search step: it finds passages related to the question and gives them to the model as context. The model is then asked to answer from those passages.

The approach has two phases. Indexing prepares the documents once, ahead of time. Querying runs each time someone asks a question.

Indexing: Preparing Documents

Indexing begins by extracting the text of each document. Long text is then split into smaller chunks, because a search works best when each result covers one idea. Overlap between neighbouring chunks keeps a sentence that crosses a boundary whole.

An embedding model converts each chunk into a vector, a list of numbers that captures its meaning. Chunks with similar meanings get vectors that are close together. The vectors are stored in a vector index, which can find the nearest ones quickly. Saving the index means the slow embedding work doesn't have to be repeated at every start.

Querying: Finding Relevant Passages

To answer a question, the system embeds the question with the same embedding model and searches the index for the nearest chunks. Plain similarity search can return several chunks that say nearly the same thing.

Maximal marginal relevance, or MMR, reduces that repetition. It first fetches a larger set of candidates, then picks results one at a time, balancing closeness to the question against similarity to the results it has already picked. A lambda value of 1 considers only closeness to the question, and lower values give more weight to variety.

Generating and Checking the Answer

The selected passages and the question are combined into a prompt, which instructs the model to answer from the passages and to say so when they don't contain the answer. Grounding like this reduces invented answers but doesn't eliminate them.

Showing the retrieved passages beside the answer lets a reader check the answer against its sources. A passage's distance score shows how similar it is to the question. It doesn't show that the passage is correct or that the answer used it faithfully. Reading the source remains the reliable way to verify a claim.